Louis Tocquec

PhD student in applied mathematics

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Interests: optimal transport, convex analysis, calculus of variations, geometric measure theory

Education

2024- **PhD in applied mathematics**, *Université Paris-Saclay and Inria*
Thesis: *On entropic optimal transport and some generalizations*
Advisors: *Luca Nenna and Paul Pegon*
Univ. Paris-Saclay: *Laboratoire de mathématiques d'Orsay*
Inria: *MokaPlan and ParMa*

2024 **MSc in applied mathematics**, *École Normale Supérieure Paris-Saclay*
Mathématiques Vision Apprentissage (MVA)
Thesis: *An ODE characterization of entropic optimal transport*

2022 **BSc in mathematics**, *Université Paris-Dauphine*
Advanced mathematics track

Research

Preprints

1. **First-order Γ-expansion for entropic optimal transport with non-degenerate costs**, *with Luca Nenna, Brendan Pass and Paul Pegon*, 2026.
HAL: https://hal.science/hal-05774627v1
2. **On convergence rate of regularized unbalanced optimal transport: the discrete case**, *with Luca Nenna and Paul Pegon*, 2025.
HAL: https://inria.hal.science/hal-05157120v2

Talks at conferences and seminars

Nov. 2026 **Seminario Imc**, PUC de Chile
Oct. 2026 **Seminario de Análisis y Geometría**, PUC de Chile
Sep. 2026 **Séminaire GOTA**, Fondation des Treilles
Mar. 2026 **SMAI-MODE**, Nice
Feb. 2026 **Séminaire des doctorants du LMO**, Institut de Mathématiques d'Orsay
Nov. 2025 **MokaMeeting**, Univ. Paris-Dauphine
Nov. 2025 **PGMO days**, EDF Lab Paris-Saclay
Oct. 2025 **GT ODE-PDE-ML**, Institut de Mathématiques d'Orsay
May. 2025 **MoKarma day**, Inria Paris
Oct. 2024 **MokaMeeting**, Inria Paris

Responsibilities

2025-2026 Organizer of the monthly seminar of the Inria MokaPlan team: "MokaMeeting".

Teaching Experience

2025-2026 Calcul numérique, Univ. Paris-Saclay, L2 Maths
Calcul scientifique avec Python, Univ. Paris-Saclay, L1 Maths
Équations différentielles, Univ. Paris-Saclay, L2 Physics
2024-2025 Calcul numérique, Univ. Paris-Saclay, L2 Maths

Languages

French Native
English Fluent
Spanish Beginner

Computer skills

Python good command
LaTeX good command
R basic knowledge
HTML basic knowledge

Miscellaneous

Licences permis B (driving licence), permis côtier (boat licence)
Sport football (competition), running (competition), tennis
Music classical guitar (15 years of practice)